

Matthew Robert Ballard

CONTACT INFORMATION	Email: mrb@mattrobball.com Webpage: www.matthewrobertballard.com
CITIZENSHIP	Born January 3, 1980 in Lansing, Michigan, United States.
INTERESTS	Categorical structures related to algebraic and arithmetic geometry, noncommutative algebra, and mirror symmetry. Formalization of mathematics.
POSITIONS	Institute for Computer-Aided Reasoning in Mathematics, Pittsburgh, Pennsylvania USA Associate Director for Scientific Activities August 2025 - Current University of South Carolina, Columbia, South Carolina USA Professor (with tenure) January 2022 - Current Simons-Laufer Mathematical Sciences Institute, Berkeley, California USA Research Professor January 2024 - May 2024 University of Michigan, Ann Arbor, Michigan USA Visiting Scholar January 2021 - May 2021 University of South Carolina, Columbia, South Carolina USA Associate Professor (with tenure) January 2018 - December 2021 Institute for Advanced Study, Princeton, New Jersey, USA Member September 2016 - July 2017 University of South Carolina, Columbia, South Carolina USA Assistant Professor August 2013 - December 2017 University of Vienna, Vienna, Austria Senior Researcher July 2012 - July 2013 University of Wisconsin, Madison, Wisconsin USA Van Vleck Visiting Assistant Professor September 2011 - May 2012 University of Pennsylvania, Philadelphia, Pennsylvania USA Postdoctoral Researcher August 2008 - May 2011
EDUCATION	University of Washington, Seattle, Washington USA Ph.D., Mathematics. June 2008. Advisor: Charles Doran. Thesis title: <i>Derived categories of quasi-projective schemes</i>. California Institute of Technology, Pasadena, California USA B.S. with honor, Mathematics. June 2002.

AWARDS

Fellow of the American Mathematical Society. Class of 2025.
 University of South Carolina Garnet Apple Award for Teaching Innovation. 2022.
 University of South Carolina College of Arts & Sciences Teaching Incubator Fellow. 2019
 University of South Carolina Breakthrough Star. 2018.

PAPERS AND PREPRINTS

Articles appear at https://arxiv.org/a/ballard_m_1.html

With Nitin Chidambaram and David Favero. *Kernels for stratified Mukai flops via the Q -construction*. In preparation.

With Alicia Lamarche. *Detecting rational points with étale exceptional collections*. In preparation.

With Christine Berkesch, Michael K. Brown, Lauren Cranton Heller, Daniel Erman, David Favero, Sheel Ganatra, Andrew Hanlon, Jesse Huang. *King's Conjecture and the Cox category*. arXiv:2501.00130. Submitted.

With Alexander Duncan, Alicia Lamarche, and Patrick McFaddin. *Consequences of the existence of exceptional collections in arithmetic and rationality*. arXiv:2009.10175. Submitted.

With Colin Diemer and David Favero. *Kernels from compactifications*. arXiv:1710.01418. Submitted.

With Srikanth B. Iyengar, Pat Lank, Alapan Mukhopadhyay, Josh Pollitz. *High Frobenius pushforwards generate the bounded derived category*. arXiv:2303.18085. Forum of Mathematics, Sigma 14 (2026) e12.

With Anne Baanen, Johan Commelin, Bryan Gin-gé Chen, Michael Rothgang, Damiano Testa. *Growing Mathlib: maintenance of a large scale mathematical library*. arXiv:2508.21593. Lecture Notes in Artificial Intelligence (CICM 2025).

With Alexander Duncan, Alicia Lamarche, and Patrick McFaddin. *Separable algebras and coflasque resolutions*. Advances in Mathematics. 444 (2024) 109596.

With Alexander Duncan and Patrick McFaddin. *Derived categories of centrally-symmetric smooth toric Fano varieties*. Mathematische Nachrichten. (2) 295 (2022), pp. 218–241.

With Blake Farman. *A category of kernels for noncommutative projective schemes*. Journal of Noncommutative Geometry. 15 (2021), 1129–1180.

With Nitin Chidambaram, David Favero, Patrick McFaddin, and Robert Vandermolen. *Kernels for Grassmann flops*. Journal de Mathématiques Pures et Appliquées. (9) 147 (2021), pp. 29–59.

With Alexander Duncan, Patrick McFaddin. *The toric Frobenius morphism and a conjecture of Orlov*. European Journal of Mathematics. 5 (2019), no. 3, 640–645.

With Alexander Duncan, Patrick McFaddin. *On derived categories of arithmetic toric varieties*. Annals of K-Theory 4 (2019), no. 2, 211–242.

With David Favero and Ludmil Katzarkov. *Variation of Geometric Invariant Theory quotients and derived categories*. Journal für die reine und angewandte Mathematik (Crelle). Issue 746 (2019) 235–304.

With Dragos Deliu, David Favero, M. Umut Isik, and Ludmil Katzarkov. *On the derived categories of degree d hypersurface fibrations*. Mathematische Annalen 371 (2018), no. 1–2, 337–370.

With Dragos Deliu, David Favero, M. Umut Isik, and Ludmil Katzarkov. *Homological Projective Duality via variation of Geometric Invariant Theory quotients*. Journal of the European Mathematical Society. Volume 19 Issue 14 (2017) 1127–1158.

Wall crossing for derived categories of moduli spaces of sheaves on rational surfaces. Algebraic

Geometry 4 (3) (2017) 263–280.

With Dragos Deliu, David Favero, M. Umut Isik, and Ludmil Katzarkov. *Resolutions in factorization categories*. Advances in Mathematics. 295 (2016) 195–249.

With Colin Diemer, David Favero, Ludmil Katzarkov, and Gabriel Kerr. *The Mori Program and non-Fano toric Homological Mirror Symmetry*. Transactions of the AMS. 367 (2015) 8933–8974.

With David Favero and Ludmil Katzarkov. *A category of kernels for equivariant factorizations and its implications for Hodge theory, I*. Publications mathématiques de l’IHÉS 120 (2014), no. 1, 1–111.

With David Favero and Ludmil Katzarkov. *A category of kernels for equivariant factorizations, II: further implications*. Journal de Mathématiques Pures et Appliquées 102 (2014), no. 4, 702–757.

With David Favero and Ludmil Katzarkov. *Orlov spectra: bounds and gaps*. Inventiones Mathematicae 189 (2012), no. 2, 359–430.

With David Favero. *Hochschild dimensions of tilting complexes*. International Mathematical Research Notices 2012 (2012), no. 11, 2607–2645.

Derived categories of sheaves on singular schemes with an application to reconstruction. Advances in Mathematics 227 (2011), no. 2, 895–919.

Equivalences of derived categories of sheaves on quasi-projective schemes. arXiv:0905.3148.

Sheaves on local Calabi-Yau varieties. arXiv:0801.3499.

Meet homological mirror symmetry in Modular Forms and String Duality. Fields Institute Communications, 54, AMS, Providence, RI, 2008.

Derived categories of sheaves on quasi-projective schemes. Thesis. 2008.

BOOKS

Superschool on derived categories and D-branes. Edmonton, Canada, July 17–23, 2016. Lectures from the PIMS Superschool. Edited with Charles Doran, David Favero and Eric Sharpe. Springer Proceedings in Mathematics & Statistics, 240. Springer, Cham, 2018.

FUNDING

National Science Foundation Mathematical Sciences Research Institute, co-PI DMS-2425401. \$6,600,000 (total). 2025–2028.

Renaissance Philanthropy AI for Math Grant, PI \$191,453. 2025–2027.

National Science Foundation Focused Research Group, PI DMS-2412040. \$137,739. 2025–2028.

National Science Foundation Standard Grant, PI DMS-2302263. \$217,590. 2023–2026.

Simons Foundation Collaboration Grant, PI. \$42,000. 2020–2025.

University of South Carolina ASPIRE - I, Tracks 3 and 4. \$15,000. 2020–2022.

Fields Institute, co-PI/organizer. \$475,000 CAD. 2019

Southeastern Conference Visiting Faculty Travel Award. \$1,800. 2019.

University of South Carolina College of Arts & Sciences Travel Grant. \$456. 2019.

National Science Foundation Standard Grant, PI DMS-1501813. \$140,000. 2015–2019.

Pacific Institute for the Mathematical Sciences, co-PI/organizer. \$10,000 CAD. 2016.

Research in Pairs – Scheme 4 London Mathematical Society, co-PI. \$1,800. 2016.

Simons Foundation Collaboration Grant, PI. \$35,000. 2014–2015.

National Science Foundation Standard Grant, co-PI/organizer DMS-1343512. \$14,620. 2013.

LECTURES (PAST 5 YEARS)

TBD. Colloquium. University of Oregon. March 2026.

The evolving capabilities of AI tools to do mathematics: stories from the frontlines. Summer

Mathematics Programs Consortium 2025 Conference. Atlanta. November 2025.

Trust and Proof in the Age of AI. Colloquium. University of Michigan, Ann Arbor. November 2025.

The king is dead. Long live the king. Workshop on Geometry. University of North Carolina, Chapel Hill. April 2025.

Mathematics, Formalization, and Computation in Lean. Formalization Colloquium. Duke University. March 2025.

Window kernels. SLMATH NAG Equivariant Derived Categories Seminar. Simons-Laufer Mathematical Sciences Institute. Berkeley. May 2024.

Hello Lean. Seminar. Simons-Laufer Mathematical Sciences Institute. Berkeley. April 2024.

Some new thoughts on old flops. Seminar on Geometry and Arithmetic. University of California, Santa Barbara. March 2024.

How complex are modules over quotients of polynomial rings? Colloquium. University of California, Santa Cruz. February 2024.

Generators for derived categories in positive characteristic. Introductory Workshop for the Noncommutative Algebraic Geometry Program. Simons-Laufer Mathematical Sciences Institute. Berkeley. January 2024.

Generation in prime characteristic. Algebraic Geometry Seminar. University of Minnesota. November 2023.

Rationality and decompositions of the derived category. Special session on Brauer Groups in Algebraic Geometry and Arithmetic. Spring Sectional Meeting of the AMS. University of Cincinnati. April 2023.

How complex are modules over quotients of polynomial rings? Colloquium. Auburn University. November 2022.

Generation in prime characteristic/A GUT for flops. Derived Categories, Arithmetic, and Reconstruction in Algebraic Geometry. Banff International Research Station. July 2022.

Generation in characteristic p . Conference on Homological Mirror Symmetry. Institute for Mathematical Sciences of the Americas. April 2022.

Kernels from flips – stratified Mukai flops. Zoom Algebraic Geometry Seminar. Online. December 2021.

Exceptional collections and rationality. Algebraic Geometry Seminar. University of Warwick. June 2021.

Exceptional collections and rationality. Algebraic Geometry Seminar. University of Michigan, Ann Arbor. March 2021.

Some comments on semi-orthogonal decompositions from quantum spectra. Workshop on Mirror Symmetry. Institute of the Mathematical Sciences of the Americas. Miami. January 2021.

Rationality and exceptional collections for toric varieties. Hodge Theory and Rationality. Institute of the Mathematical Sciences of the Americas. Miami. October 2020.

TEACHING

University of South Carolina, Columbia, South Carolina USA

Instructor

Solely responsible for lectures, exams, homework assignments, and grades.

- Math 141 Calculus I Fall 2014, Fall 2018, Spring 2019, Fall 2019x2
- Math 141 Honors Calculus I Fall 2015
- Math 142 Calculus II Fall 2013, Spring 2025
- Math 241 Calculus III Fall 2023
- Math 241 Honors Calculus III Fall 2017

• Math 300 Honors Transition to Advanced Mathematics	Fall 2020, Fall 2022
• Math 344 Honors Applied Linear Algebra	Spring 2025
• SCHC 411 Formalization and Mathematics	Spring 2022
• Math 544 Linear Algebra	Fall 2013
• Math 544 Honors Linear Algebra	Fall 2021
• Math 546 Algebraic Structures I	Spring 2014
• CSCE 557/Math 587 Introduction to Cryptography	Fall 2022, Fall 2024
• Math 701 Foundations of Algebra I	Fall 2015
• Math 702 Foundations of Algebra II	Spring 2016
• Math 732 Algebraic Topology I	Fall 2014
• Math 733 Algebraic Topology II	Spring 2015
• Math 737 Introduction to Complex Geometry I	Fall 2017
• Math 738 Derived Categories I	Fall 2021
• Math 748 Derived Categories II	Spring 2022
• Math 768 Formalization of Mathematics	Spring 2023

University of Wisconsin, Madison, Wisconsin USA

Instructor

Solely responsible for lectures, exams, homework assignments, and grades.

• Math 475 Introduction to Combinatorics	Fall 2011
• Math 541 Modern Algebra I	Spring 2012

University of Pennsylvania, Philadelphia, Pennsylvania USA

Instructor

Solely responsible for lectures, exams, homework assignments, and grades.

• Math 104 Calculus II	Fall 2010
• Math 114 Calculus III	Spring 2009
• Math 505 Graduate Proseminar in Mathematics	Spring 2011
• Math 622 Complex Algebraic Geometry I	Fall 2009
• Math 623 Complex Algebraic Geometry II	Spring 2010

University of Washington, Seattle, Washington USA

Instructor

Solely responsible for lectures, exams, homework assignments, and grades.

• Math 126 Calculus and Analytic Geometry III	Summer 2004
• Math 307 Introduction to Differential Equations	Summer 2005
• Math 308 Introduction to Linear Algebra	Summer 2007
• Math 309 Linear Analysis	Summer 2008

MENTORING

- Uttaran Dutta, Ph.D., USC, Mathematics, Expected 2027.
- Anirban Bhaduri, Ph.D., USC, Mathematics, Expected 2026.
- Patrick Lank, Ph.D., USC, Mathematics, July 2024. Currently, postdoctoral researcher at Università degli Studi di Milano.
- Keller VandeBogert, Ph.D., USC, Mathematics, May 2021. (co-advised with Andrew Kustin). Currently tenure-track assistant professor at the University of Kentucky.
- Alicia Lamarche. Ph.D., USC, Mathematics, May 2020. Currently Tsinghua-YMSC-Imperial College postdoc in algebraic geometry at the Yau Mathematical Sciences Center.
- Robert Vandermolen. Ph.D., USC, Mathematics, May 2020. Currently tenure track assistant professor at Saint Mary-of-the-Woods College.
- Patrick McFaddin. Postdoc, USC. 2016 – 2019. Currently tenured associate professor at Fordham University.
- Jessisa Otis, M.S., USC, Mathematics. 2019.
- Blake Farman. Ph.D., USC, Mathematics. 2018. Currently tenure-track assistant professor at

Louisiana Tech University.

- Ross Berkowitz. Masters, UPenn, Mathematics. May 2011.

PROFESSIONAL
MEMBERSHIP

Member of the American Mathematical Society

REVIEWER

Reviewer for Advances in Mathematics, Advances in Theoretical and Mathematical Physics, Algebraic Geometry, Applied Categorical Structures, Annals of K-theory, Banff International Research Station, Cambridge Mathematical Journal, Communications in Algebra, Compositio Mathematica, Duke Mathematical Journal, European Research Council, Inventiones Mathematicae, Journal of Algebra, Journal of Algebra and its Applications, Journal of Algebraic Geometry, Journal of Differential Geometry, Journal de Mathématiques Pures et Appliquées, Journal of Pure and Applied Algebra, Journal für die Reine und Angewandte Mathematik, Mathematical Research Letters, Mathematische Annalen, Mathematische Zeitschrift, Mathlib, Michigan Journal of Mathematics, National Security Agency, National Science Foundation, Natural Sciences and Engineering Research Council of Canada, Proceedings of the AMS, Proceedings of the Fields' Institute, Proceedings of String-Math, and Transactions of the AMS.

SERVICE

- Maintainer, Mathlib, library of formalized mathematics in the Lean programming language. 2023–2025.
- Calculus Director, Department of Mathematics. 2023–2024. USC.
- Co-chair of Service Courses Committee, Department of Mathematics. 2023–2024. USC.
- Undergraduate mathematics advisor 2013–2015, 2021–2024. USC.
- Education Committee, Department of Mathematics. 2021–2022. USC.
- Carolina Top Scholar Selection Reviewer. 2020. USC.
- Course coordinator for Calculus I 2018–2020. USC.
- Chair of Special Committee on Calculus Sequence. 2018. USC.
- co-Editor of proceedings of Superschool on Derived Categories and D-branes. 2018.
- Graduate Advisory Committee. 2015–2016, 2017–2019. USC.
- Mathematics Self-study Committee 2017–2018. USC.
- Faculty liaison for IIME Honor Society and Gamecock Math Club 2013–2016. USC.
- Hiring committee 2013–2016. USC.
- Faculty senator 2013–2016. USC.
- Graduate Admission Committee 2009–2011. UPenn.
- Member of Comprehensive Exam committee for Bailey Heath, 2021; Jonathan Smith, 2021; Jaree Hudson, 2015; USC Masters Thesis committee for Marvin Jones, 2014; USC Ph.D. committee for Richard Oh, 2014.

ORGANIZING

- AMS Mathematical Research Community on Derived Categories, Arithmetic, and Geometry. July 2023.
- Special Session on Derived Categories and Birational Geometry at joint Meeting of the American Mathematical Society, Société Mathématique de France, and European Mathematical Society. July 2022.
- Birational, Categorical, and Derived Algebraic Geometry. Banff International Research Station. November 2021.
- Special Session on Mathematical-Physical Aspects of Toric and Tropical Geometry at the Mathematical Congress of the Americas. July 2021.
- USC Algebra, Geometry, and Number Theory Seminar 2013–2021.
- Session on Derived Categories and (Non)Commutative Algebraic Geometry. Winter Meeting of the Canadian Mathematical Society. December 2020.
- Birational, Categorical, and Derived Algebraic Geometry. Banff International Research Station. November 2020.
- Thematic Program on Homological Mirror Symmetry. Fields Institute. August–December 2019.
- USC Graduate Colloquium 2015–2016.

- Superschool on Derived Categories and D-branes. July 2016.
- Banff International Research Station Workshop on Homological Mirror Geometry. March 2016.
- AMS Special Session on Interactions between Algebraic and Tropical Geometry. March 2016.
- AMS Special Session on Mirror Symmetry. Southeast Section. November 2014.
- Commutative Algebra – Algebraic Geometry in the Southeast, November 2013.
- Geometry of D-branes thematic period, Erwin Schrödinger Institute, April – July 2013.
- Birational Geometry and Derived Categories conference, University of Vienna, August 2012.